

Vinay Jampana

Senior Software Engineer – AI & Full-Stack

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Summary

AI full-stack engineer with 5 years across Python, Node.js, NestJS, AWS, TypeScript, and React. I ship production AI agents end to end, from architecture to deployment, with evals, tracing, and monitoring built in. Experienced in RBAC, RAG, and multi-agent orchestration, and in leading small teams to deliver reliable, maintainable systems.

Skills

Backend: Node.js, Python, TypeScript, NestJS, FastAPI, PostgreSQL

Agentic AI: LLM agents, LangGraph, RAG, LiteLLM, OpenSearch, Langfuse

Frontend: React, Next.js

Cloud: AWS, Docker, GitHub Actions

Experience

Zotok.ai – Senior Software Engineer (AI & Full-Stack)

May 2023 – Present

Stack: Node.js, Python, NestJS, LangGraph, PostgreSQL, OpenSearch, React, AWS

AI engineering

- Redesigned the agent (**LangGraph**, NestJS) from one large block into seven independent, testable steps on a visual workflow builder over 50 functions, so customer-specific changes need no code change or redeployment
- Designed the **multi-agent** flow of the WhatsApp sales agent (LangGraph): each message goes to either the service agent or the payment agent, and a **guardrail** classifier filters spam, auto-replies and off-topic messages
- Built an **LLM evals** layer that works as observability for the agent: 64 scenarios run twice a day against the live flow, with Teams alerts, so a change anywhere in the stack shows up as a score change within hours
- Built the agent's **RAG** retrieval (OpenSearch, embeddings via a LiteLLM gateway) with **multi-tenant** isolation, so each customer gets answers only from their own documents
- Built a product matching engine for shorthand order text over 15,242 production aliases: correct first match went from 69% to **78%** in testing, and from 81% to **95%** on real user messages
- Built a **multimodal** image-to-order pipeline: photos of handwritten order sheets are converted to carts using OCR and product matching, with **88%** line accuracy on a 378-line test set
- Designed **agent memory** for the order agent: a per-customer phrase-to-product memory with a workspace-wide fallback, passed to the model as evidence and saved only after the customer confirms a choice
- Integrated an **LLM gateway** (LiteLLM) with separate keys per workspace and spend tags per feature, so model cost can be tracked customer-wise and feature-wise

Backend engineering

- Designed and built a **WhatsApp workflow** (agent, NestJS backend, React web app) that replaced a plant's offline pour and dispatch forms; over 3,800 pour cards and 1,900 dispatches captured since launch
- Designed **human-in-the-loop** handover for the WhatsApp assistant: messages the bot can handle are answered at once, others wait for a human, and each conversation is processed in order
- Built the backend service for the agent (NestJS, PostgreSQL): over 50 functions sharing one common contract, and product search made **4x faster** (875 ms to 228 ms per line)

Frontend engineering

- Built dynamic WhatsApp template buttons: five button types are resolved at send time into **deep links**, so each buyer lands on the right page, from the template builder to the landing page
- Reduced first page load from 15–30 seconds to **1.5 seconds** by removing unused mobile code, rebuilding the Webpack config and adding lazy loading; builds also became 30–40% faster

Sumeru Software – Senior Developer

Aug 2022 – Jan 2023

- Built a role-based CRM from scratch (React, Redux Toolkit, Node.js) with permission-based screens and modular dashboards; owned it from kickoff to production and delivered the MVP in **4 months** with a team of 4

Aerchain – Software Development Engineer

Intern Jul 2019 – Jul 2020; SDE 1 Apr 2021 – Jan 2022

- Built a real-time auction module for a B2B procurement platform with an **optimistic UI** that syncs with the server, and barcode-scan inventory management as an **offline-first** PWA

Education

B.E. (Hons) Mechanical Engineering, BITS Pilani

2016 – 2020